

EVENTFLOW

EF-DOC-005

Database Design Specification

Version 1.0 | Approved Baseline

Sprint 3 - Database Engineering

Logical Data Model Draft Baseline 1 - Accepted

Physical Data Model Draft Baseline 1 - Integrity Gate Passed

Every Guest Matters. Every Event Flows.

Document Control

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Related	EF-DOC-000, 001, 002, 003, 004; ADR-013 through ADR-021

1. Purpose

This document consolidates the reviewed EventFlow v1.x database architecture into an implementation-grade relational baseline. It defines database standards, logical ownership, physical tables, data types, indexes, foreign keys, migration rules and integrity invariants. The current status is Approved Baseline; the logical and physical baselines have passed their architecture review gates and are accepted for Sprint 3.

2. Integrity Gate Result

Physical Data Model Draft Baseline 1: PASS

The final gate validated entity completeness, multi-event isolation, event authorization, referential integrity, seating/grouping, check-in history, communications, webhook idempotency, imports, migration governance, UTC/timezone rules, privacy boundaries and end-to-end Event lifecycle behavior.

3. Approved Database Standards

ID	Standard	Accepted Rule
ADR-021	Event-scoped domain model	One installation manages multiple independent Events; Invitations and Attendees belong to exactly one Event; no global Person model in v1.x.
DB-001	Primary keys	BIGINT UNSIGNED AUTO_INCREMENT internal IDs; public/security references use opaque tokens/identifiers.
DB-002	Referential integrity	Database-enforced foreign keys for core EventFlow relations; RESTRICT/NO ACTION by default; cascades only for clearly bounded technical relationships.
DB-003	Naming	Lowercase snake_case; active WordPress prefix + eventflow_ namespace; explicit entity-qualified key names.
DB-004	Time	UTC DATETIME storage; each Event stores an IANA timezone for local display/scheduling.
DB-005	Retention	Distinguish lifecycle completion, archive, soft deletion and explicit purge/anonymization.
DB-006	State	Explicit string state fields; no MySQL ENUM for business lifecycles; transitions enforced by domain services.
DB-007	Database platform	MySQL 8.0+ or MariaDB 10.11+, InnoDB, utf8mb4, transactions, foreign keys, portable logical JSON handling.
PDM-R01	Composite Event-scoped FKs	Use composite event_id + entity_id foreign keys wherever practical to prevent cross-Event references at SQL level.
PDM-017A	Provider webhook idempotency	Every provider event has mandatory EventFlow event_dedupe_key with unique provider+dedupe constraint.

4. Logical Model and ERDs

Event is the operational ownership boundary. Venue is reusable master data. Invitation is an entitlement/container; Attendee is the person. Seating and check-in remain separate operational entities. Communications preserve campaign, message, transmission attempt and provider-event history.

Figure 1 - Core Event, Invitation, Attendee, Seating and Check-in ERD

Figure 2 - Communications, Import, Audit and Migration ERD

5. Physical Table Catalog

PDM	Table	Purpose
PDM-002	eventflow_venues	Reusable venue master data shared by multiple Events.
PDM-001	eventflow_events	Root EventFlow business table and Event ownership boundary.
PDM-003	eventflow_event_configurations	One typed configuration row per Event for branding and event behavior.
PDM-004	eventflow_invitations	Event-scoped invitation entitlement, primary delivery contact, capacity and secure guest access.
PDM-005	eventflow_attendees	One real event-scoped individual linked to one Invitation.
PDM-006	eventflow_seating_groups	Event-scoped invitation/family and host-defined affinity groups used by seating recommendations.
PDM-007	eventflow_seating_group_members	Many-to-many membership between Attendees and Seating Groups.
PDM-008	eventflow_tables	Organizer-configurable event seating tables with variable capacities.
PDM-009	eventflow_seats	Optional physical seats for seat-specific Events.
PDM-010	eventflow_seating_assignments	Authoritative and historical placement of Attendees at Tables and optional Seats.
PDM-011	eventflow_checkin_stations	Optional logical check-in locations for an Event.
PDM-012	eventflow_checkins	Append-only attendee-level attendance actions and explicit reversals.
PDM-013	eventflow_communication_templates	Versioned reusable event communication content.
PDM-014	eventflow_campaigns	Event-scoped organizer communication operation and audience rule.
PDM-015	eventflow_messages	One logical communication to one recipient with rendered historical snapshot.
PDM-016	eventflow_message_delivery_attempts	Append-oriented provider transmission attempts for a Message.
PDM-017 / 017A	eventflow_provider_events	Asynchronous provider-originated communication events with mandatory EventFlow deduplication.
PDM-018	eventflow_audit_logs	Append-only business and security history for material EventFlow operations.
PDM-019	eventflow_event_venue_snapshots	Event-specific historical venue truth captured when location is operationally locked.
PDM-020	eventflow_schema_migrations	Authoritative history for schema and controlled data migrations.
PDM-021A	eventflow_import_jobs	Parent record for one staged CSV/XLSX import operation.
PDM-021B	eventflow_import_rows	Staged, normalized and validated

		source rows before production apply.
PDM-022	eventflow_event_memberships	Event-scoped authorization between WordPress users and Events.

PDM-002 - eventflow_venues

Reusable venue master data shared by multiple Events.

Column	MySQL Type	Null	Default	Purpose
venue_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
venue_name	VARCHAR(190)	NO	-	Display name
venue_status	VARCHAR(32)	NO	'active'	Lifecycle state
address_line_1	VARCHAR(190)	YES	NULL	Primary street address
address_line_2	VARCHAR(190)	YES	NULL	Suite/unit/additional address
city	VARCHAR(120)	YES	NULL	City/locality
region	VARCHAR(120)	YES	NULL	Province/state/region
postal_code	VARCHAR(32)	YES	NULL	Postal/ZIP code
country_code	CHAR(2)	YES	NULL	ISO-style two-letter country code
latitude	DECIMAL(10,7)	YES	NULL	Optional latitude
longitude	DECIMAL(10,7)	YES	NULL	Optional longitude
phone	VARCHAR(40)	YES	NULL	Venue phone
email	VARCHAR(190)	YES	NULL	Venue email
website_url	VARCHAR(500)	YES	NULL	Venue website
default_capacity	INT UNSIGNED	YES	NULL	Informational normal capacity
notes	TEXT	YES	NULL	Organizer notes
created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
updated_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC modification time
deleted_at	DATETIME	YES	NULL	Soft deletion

Indexes and uniqueness

PRIMARY KEY (venue_id)

KEY idx_venue_status (venue_status)

KEY idx_venue_location (country_code, region, city)

KEY idx_venue_name (venue_name)

KEY idx_venue_deleted (deleted_at)

PDM-001 - eventflow_events

Root EventFlow business table and Event ownership boundary.

Column	MySQL Type	Null	Default	Purpose
event_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_name	VARCHAR(190)	NO	-	Official event name
event_slug	VARCHAR(190)	NO	-	Installation-unique stable slug
event_status	VARCHAR(32)	NO	'draft'	draft/active/completed/cancelled/archived
starts_at	DATETIME	YES	NULL	UTC event start
ends_at	DATETIME	YES	NULL	UTC event end
timezone	VARCHAR(64)	NO	-	IANA timezone

venue_id	BIGINT UNSIGNED	YES	NULL	Primary reusable Venue
created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
updated_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC modification time
deleted_at	DATETIME	YES	NULL	Soft deletion

Indexes and uniqueness

PRIMARY KEY (event_id)

UNIQUE KEY uq_event_slug (event_slug)

KEY idx_event_status (event_status)

KEY idx_event_dates (starts_at, ends_at)

KEY idx_event_venue (venue_id)

KEY idx_event_deleted (deleted_at)

Relationships / foreign keys

venue_id -> eventflow_venues.venue_id ON DELETE RESTRICT ON UPDATE RESTRICT

PDM-003 - eventflow_event_configurations

One typed configuration row per Event for branding and event behavior.

Column	MySQL Type	Null	Default	Purpose
event_configuration_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	One-to-one Event
logo_media_id	BIGINT UNSIGNED	YES	NULL	WordPress attachment reference
invitation_media_id	BIGINT UNSIGNED	YES	NULL	WordPress attachment reference
primary_theme	VARCHAR(64)	YES	NULL	Theme identifier
secondary_theme	VARCHAR(64)	YES	NULL	Optional secondary theme
welcome_message	TEXT	YES	NULL	Guest landing message
confirmation_message	TEXT	YES	NULL	Post-confirmation message
surprise_notice	TEXT	YES	NULL	Optional guest notice
dress_code	VARCHAR(255)	YES	NULL	Guest-facing dress code
confirmation_opens_at	DATETIME	YES	NULL	UTC confirmation opening
confirmation_closes_at	DATETIME	YES	NULL	UTC confirmation closing
allow_guest_edits	TINYINT(1)	NO	0	Guest post-submission edits enabled
seating_mode	VARCHAR(32)	NO	'table'	table or seat
automatic_seating_enabled	TINYINT(1)	NO	0	Automatic seating recommendations enabled
default_from_name	VARCHAR(190)	YES	NULL	Email sender display name
reply_to_email	VARCHAR(190)	YES	NULL	Event-level reply-to
default_sms_sender	VARCHAR(64)	YES	NULL	SMS sender identifier
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC modification time
created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
updated_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference

Indexes and uniqueness

PRIMARY KEY (event_configuration_id)

UNIQUE KEY uq_event_configuration_event (event_id)

KEY idx_event_configuration_seating_mode (seating_mode)

Relationships / foreign keys

event_id -> eventflow_events.event_id RESTRICT

PDM-004 - eventflow_invitations

Event-scoped invitation entitlement, primary delivery contact, capacity and secure guest access.

Column	MySQL Type	Null	Default	Purpose
invitation_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
invitation_code	VARCHAR(32)	NO	-	Organizer-friendly code
primary_name	VARCHAR(190)	NO	-	Primary invitee display name
primary_email	VARCHAR(190)	YES	NULL	Delivery email
primary_email_normalized	VARCHAR(190)	YES	NULL	Normalized email
primary_phone	VARCHAR(40)	YES	NULL	Delivery phone
primary_phone_normalized	VARCHAR(32)	YES	NULL	Normalized phone
capacity	SMALLINT UNSIGNED	NO	1	Maximum active attendees
invitation_status	VARCHAR(32)	NO	'active'	Invitation lifecycle
response_status	VARCHAR(32)	NO	'pending'	pending/accepted/declined
token_lookup	BINARY(32)	NO	-	SHA-256 digest for indexed token lookup
token_version	SMALLINT UNSIGNED	NO	1	Token rotation generation
token_expires_at	DATETIME	YES	NULL	UTC optional expiry
token_revoked_at	DATETIME	YES	NULL	UTC revocation
first_accessed_at	DATETIME	YES	NULL	First successful guest access
last_accessed_at	DATETIME	YES	NULL	Latest successful guest access
submitted_at	DATETIME	YES	NULL	Latest completed confirmation submission
declined_at	DATETIME	YES	NULL	Decline time
organizer_notes	TEXT	YES	NULL	Internal notes
created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
updated_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC modification time
deleted_at	DATETIME	YES	NULL	Soft deletion

Indexes and uniqueness

PRIMARY KEY (invitation_id)

UNIQUE KEY uq_invitation_event_id (event_id, invitation_id)

UNIQUE KEY uq_invitation_event_code (event_id, invitation_code)

UNIQUE KEY uq_invitation_token_lookup (token_lookup)

KEY idx_invitation_event_status (event_id, invitation_status)

KEY idx_invitation_event_response (event_id, response_status)

KEY idx_invitation_email (event_id, primary_email_normalized)

KEY idx_invitation_phone (event_id, primary_phone_normalized)

KEY idx_invitation_submitted (event_id, submitted_at)

KEY idx_invitation_deleted (event_id, deleted_at)

Relationships / foreign keys

event_id -> eventflow_events.event_id RESTRICT

PDM-005 - eventflow_attendees

One real event-scoped individual linked to one Invitation.

Column	MySQL Type	Null	Default	Purpose
attendee_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
invitation_id	BIGINT UNSIGNED	NO	-	Parent Invitation
first_name	VARCHAR(120)	YES	NULL	Given name
last_name	VARCHAR(120)	YES	NULL	Family name
display_name	VARCHAR(190)	NO	-	Canonical display name
attendee_role	VARCHAR(32)	NO	'companion'	primary/companion/future roles
email	VARCHAR(190)	YES	NULL	Person-level email
email_normalized	VARCHAR(190)	YES	NULL	Normalized email
phone	VARCHAR(40)	YES	NULL	Person-level phone
phone_normalized	VARCHAR(32)	YES	NULL	Normalized phone
attendance_status	VARCHAR(32)	NO	'pending'	pending/confirmed/declined/cancelled
confirmed_at	DATETIME	YES	NULL	UTC confirmation time
declined_at	DATETIME	YES	NULL	UTC decline time
cancelled_at	DATETIME	YES	NULL	UTC cancellation time
dietary_requirements	TEXT	YES	NULL	Dietary requirements
accessibility_requirements	TEXT	YES	NULL	Accessibility/ accommodation requirements
organizer_notes	TEXT	YES	NULL	Internal organizer notes
created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
updated_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC modification time
deleted_at	DATETIME	YES	NULL	Soft deletion

Indexes and uniqueness

PRIMARY KEY (attendee_id)

UNIQUE KEY uq_attendee_event_id (event_id, attendee_id)

KEY idx_attendee_invitation (event_id, invitation_id)

KEY idx_attendee_event_status (event_id, attendance_status)

KEY idx_attendee_event_role (event_id, attendee_role)

KEY idx_attendee_event_name (event_id, last_name, first_name)

KEY idx_attendee_display_name (event_id, display_name)

KEY idx_attendee_email (event_id, email_normalized)

KEY idx_attendee_phone (event_id, phone_normalized)

KEY idx_attendee_deleted (event_id, deleted_at)

Relationships / foreign keys

event_id -> events RESTRICT

(event_id, invitation_id) -> invitations(event_id, invitation_id) RESTRICT

PDM-006 - eventflow_seating_groups

Event-scoped invitation/family and host-defined affinity groups used by seating recommendations.

Column	MySQL Type	Null	Default	Purpose
seating_group_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
group_name	VARCHAR(190)	NO	-	Group display name
group_category	VARCHAR(64)	NO	'custom'	family/church/school/ work/friends/association/ community/vip/custom
group_source	VARCHAR(32)	NO	'host_defined'	invitation/host_defined/ system
source_invitation_id	BIGINT UNSIGNED	YES	NULL	Invitation source when derived
constraint_level	VARCHAR(32)	NO	'preferred'	required/preferred/ informational
priority	SMALLINT UNSIGNED	NO	100	Relative recommendation priority
group_status	VARCHAR(32)	NO	'active'	Lifecycle state
organizer_notes	TEXT	YES	NULL	Internal notes
created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
updated_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC modification time
deleted_at	DATETIME	YES	NULL	Soft deletion

Indexes and uniqueness

PRIMARY KEY (seating_group_id)

UNIQUE KEY uq_seating_group_event_id (event_id, seating_group_id)

KEY idx_seating_group_event_category (event_id, group_category)

KEY idx_seating_group_event_source (event_id, group_source)

KEY idx_seating_group_invitation (event_id, source_invitation_id)

KEY idx_seating_group_constraint (event_id, constraint_level, priority)

KEY idx_seating_group_deleted (event_id, deleted_at)

Relationships / foreign keys

event_id -> events RESTRICT

(event_id, source_invitation_id) -> invitations composite RESTRICT

PDM-007 - eventflow_seating_group_members

Many-to-many membership between Attendees and Seating Groups.

Column	MySQL Type	Null	Default	Purpose
seating_group_member_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
seating_group_id	BIGINT UNSIGNED	NO	-	Group

attendee_id	BIGINT UNSIGNED	NO	-	Attendee member
membership_source	VARCHAR(32)	NO	'manual'	invitation/manual/system/import
created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
created_at	DATETIME	NO	-	UTC creation time

Indexes and uniqueness

PRIMARY KEY (seating_group_member_id)

UNIQUE KEY uq_seating_group_member (seating_group_id, attendee_id)

KEY idx_group_member_event (event_id)

KEY idx_group_member_attendee (event_id, attendee_id)

Relationships / foreign keys

event_id -> events RESTRICT

(event_id,seating_group_id) -> seating_groups composite

(event_id,attendee_id) -> attendees composite

Implementation notes

Relationship rows are removed through controlled application actions; audit history preserves changes.

PDM-008 - eventflow_tables

Organizer-configurable event seating tables with variable capacities.

Column	MySQL Type	Null	Default	Purpose
table_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
table_name	VARCHAR(190)	NO	-	Display label
table_number	INT UNSIGNED	YES	NULL	Optional numeric identity
table_type	VARCHAR(64)	NO	'standard'	standard/VIP/head/future types
capacity	SMALLINT UNSIGNED	NO	-	Maximum active occupancy
table_status	VARCHAR(32)	NO	'active'	Lifecycle state
sort_order	INT UNSIGNED	NO	100	Display order
position_x	DECIMAL(10,4)	YES	NULL	Future floor-plan X
position_y	DECIMAL(10,4)	YES	NULL	Future floor-plan Y
organizer_notes	TEXT	YES	NULL	Internal notes
created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
updated_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC modification time
deleted_at	DATETIME	YES	NULL	Soft deletion

Indexes and uniqueness

PRIMARY KEY (table_id)

UNIQUE KEY uq_table_event_id (event_id, table_id)

UNIQUE KEY uq_table_event_name (event_id, table_name)

KEY idx_table_event_status (event_id, table_status)

KEY idx_table_event_type (event_id, table_type)

KEY idx_table_event_order (event_id, sort_order)

KEY idx_table_deleted (event_id, deleted_at)

Relationships / foreign keys

event_id -> events RESTRICT

PDM-009 - eventflow_seats

Optional physical seats for seat-specific Events.

Column	MySQL Type	Null	Default	Purpose
seat_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
table_id	BIGINT UNSIGNED	NO	-	Parent Table
seat_number	SMALLINT UNSIGNED	YES	NULL	Optional numeric seat position
seat_label	VARCHAR(64)	NO	-	Human-friendly label
sort_order	SMALLINT UNSIGNED	NO	100	Display order
seat_status	VARCHAR(32)	NO	'active'	active/inactive/reserved/ blocked
is_accessible	TINYINT(1)	NO	0	Accessibility-designated seat
organizer_notes	TEXT	YES	NULL	Internal notes
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC modification time
deleted_at	DATETIME	YES	NULL	Soft deletion

Indexes and uniqueness

PRIMARY KEY (seat_id)

UNIQUE KEY uq_seat_event_id (event_id, seat_id)

UNIQUE KEY uq_seat_event_table_id (event_id, table_id, seat_id)

UNIQUE KEY uq_seat_table_label (table_id, seat_label)

UNIQUE KEY uq_seat_table_number (table_id, seat_number)

KEY idx_seat_table_status (event_id, table_id, seat_status)

KEY idx_seat_event_accessible (event_id, is_accessible)

KEY idx_seat_deleted (event_id, deleted_at)

Relationships / foreign keys

event_id -> events RESTRICT

(event_id,table_id) -> tables composite RESTRICT

PDM-010 - eventflow_seating_assignments

Authoritative and historical placement of Attendees at Tables and optional Seats.

Column	MySQL Type	Null	Default	Purpose
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seating_assignment_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
attendee_id	BIGINT UNSIGNED	NO	-	Assigned Attendee
table_id	BIGINT UNSIGNED	NO	-	Assigned Table
seat_id	BIGINT UNSIGNED	YES	NULL	Optional assigned Seat
assignment_source	VARCHAR(32)	NO	'manual'	manual/automatic/imported/system
assignment_status	VARCHAR(32)	NO	'active'	active/released/superseded
has_group_override	TINYINT(1)	NO	0	Host intentionally overrode grouping
override_reason	VARCHAR(500)	YES	NULL	Override reason
assignment_reason	VARCHAR(500)	YES	NULL	Placement rationale
assigned_at	DATETIME	NO	-	UTC assignment time
released_at	DATETIME	YES	NULL	UTC release time
created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
updated_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC modification time

Indexes and uniqueness

PRIMARY KEY (seating_assignment_id)

KEY idx_assignment_attendee (event_id, attendee_id, assignment_status)

KEY idx_assignment_table (event_id, table_id, assignment_status)

KEY idx_assignment_seat (event_id, seat_id, assignment_status)

KEY idx_assignment_source (event_id, assignment_source)

KEY idx_assignment_active_tables (event_id, assignment_status, table_id)

Relationships / foreign keys

event_id -> events

(event_id,attendee_id)->attendees

(event_id,table_id)->tables

(event_id,table_id,seat_id)->seats

Implementation notes

One active placement per Attendee and one active attendee per Seat are transactionally enforced by SeatingService.

PDM-011 - eventflow_checkin_stations

Optional logical check-in locations for an Event.

Column	MySQL Type	Null	Default	Purpose
station_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
station_name	VARCHAR(190)	NO	-	Display name
station_code	VARCHAR(32)	YES	NULL	Optional short code
station_type	VARCHAR(32)	NO	'staffed'	staffed/self_service/future
station_status	VARCHAR(32)	NO	'active'	Lifecycle state
sort_order	INT UNSIGNED	NO	100	Display order
organizer_notes	TEXT	YES	NULL	Internal notes
created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
updated_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference

created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC modification time
deleted_at	DATETIME	YES	NULL	Soft deletion

Indexes and uniqueness

PRIMARY KEY (station_id)

UNIQUE KEY uq_station_event_id (event_id, station_id)

UNIQUE KEY uq_station_event_name (event_id, station_name)

UNIQUE KEY uq_station_event_code (event_id, station_code)

KEY idx_station_event_status (event_id, station_status)

KEY idx_station_event_type (event_id, station_type)

KEY idx_station_event_order (event_id, sort_order)

KEY idx_station_deleted (event_id, deleted_at)

Relationships / foreign keys

event_id -> events RESTRICT

PDM-012 - eventflow_checkins

Append-only attendee-level attendance actions and explicit reversals.

Column	MySQL Type	Null	Default	Purpose
checkin_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
attendee_id	BIGINT UNSIGNED	NO	-	Attendee
station_id	BIGINT UNSIGNED	YES	NULL	Optional Station
action_type	VARCHAR(32)	NO	'check_in'	check_in/reversal/future actions
checkin_method	VARCHAR(32)	NO	'manual'	manual/search/guest_list/qr_code
reversal_of_checkin_id	BIGINT UNSIGNED	YES	NULL	Original action being reversed
performed_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress operator
operation_id	CHAR(36)	YES	NULL	Groups multi-attendee operation
reason	VARCHAR(500)	YES	NULL	Correction/reversal reason
notes	TEXT	YES	NULL	Operational notes
occurred_at	DATETIME	NO	-	UTC business occurrence
created_at	DATETIME	NO	-	UTC persistence time

Indexes and uniqueness

PRIMARY KEY (checkin_id)

UNIQUE KEY uq_checkin_event_id (event_id, checkin_id)

KEY idx_checkin_event_attendee (event_id, attendee_id, occurred_at)

KEY idx_checkin_event_time (event_id, occurred_at)

KEY idx_checkin_station_time (event_id, station_id, occurred_at)

KEY idx_checkin_event_action (event_id, action_type, occurred_at)

KEY idx_checkin_operation (operation_id)

KEY idx_checkin_reversal (event_id, reversal_of_checkin_id)

Relationships / foreign keys

event_id -> events

(event_id,attendee_id)->attendees

(event_id,station_id)->checkin_stations

(event_id,reversal_of_checkin_id)->checkins

PDM-013 - eventflow_communication_templates

Versioned reusable event communication content.

Column	MySQL Type	Null	Default	Purpose
template_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
template_key	VARCHAR(100)	NO	-	Stable logical template identifier
template_name	VARCHAR(190)	NO	-	Organizer-facing name
channel	VARCHAR(32)	NO	-	email/sms/future
template_type	VARCHAR(64)	NO	'general'	invitation/reminder/update/etc.
template_status	VARCHAR(32)	NO	'draft'	draft/published/archived
version_number	INT UNSIGNED	NO	1	Content version
subject_template	VARCHAR(500)	YES	NULL	Email subject source
body_template	LONGTEXT	NO	-	Body source
plain_text_template	LONGTEXT	YES	NULL	Plain-text fallback
allowed_merge_fields	JSON	YES	NULL	Controlled merge field metadata
published_at	DATETIME	YES	NULL	UTC publication
archived_at	DATETIME	YES	NULL	UTC archive
created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
updated_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC modification time
deleted_at	DATETIME	YES	NULL	Soft deletion

Indexes and uniqueness

PRIMARY KEY (template_id)

UNIQUE KEY uq_template_event_id (event_id, template_id)

UNIQUE KEY uq_template_version (event_id, template_key, channel, version_number)

KEY idx_template_event_status (event_id, template_status)

KEY idx_template_event_channel (event_id, channel)

KEY idx_template_event_type (event_id, template_type)

KEY idx_template_key (event_id, template_key)

KEY idx_template_deleted (event_id, deleted_at)

Relationships / foreign keys

event_id -> events RESTRICT

Implementation notes

Published versions are immutable; changes create a new version.

PDM-014 - eventflow_campaigns

Event-scoped organizer communication operation and audience rule.

Column	MySQL Type	Null	Default	Purpose
campaign_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
template_id	BIGINT UNSIGNED	YES	NULL	Selected template version
campaign_name	VARCHAR(190)	NO	-	Organizer-facing name
channel	VARCHAR(32)	NO	-	Delivery channel
campaign_type	VARCHAR(64)	NO	'general'	Business purpose
campaign_status	VARCHAR(32)	NO	'draft'	Campaign lifecycle
audience_definition	JSON	NO	-	Structured selection rule
audience_snapshot_count	INT UNSIGNED	YES	NULL	Resolved recipient count
scheduled_at	DATETIME	YES	NULL	UTC schedule
started_at	DATETIME	YES	NULL	UTC start
completed_at	DATETIME	YES	NULL	UTC completion
cancelled_at	DATETIME	YES	NULL	UTC cancellation
queued_count	INT UNSIGNED	NO	0	Derived summary
provider_accepted_count	INT UNSIGNED	NO	0	Derived summary
delivered_count	INT UNSIGNED	NO	0	Derived summary
failed_count	INT UNSIGNED	NO	0	Derived summary
bounced_count	INT UNSIGNED	NO	0	Derived summary
created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
updated_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC modification time
deleted_at	DATETIME	YES	NULL	Soft deletion

Indexes and uniqueness

PRIMARY KEY (campaign_id)

UNIQUE KEY uq_campaign_event_id (event_id, campaign_id)

KEY idx_campaign_event_status (event_id, campaign_status)

KEY idx_campaign_event_channel (event_id, channel)

KEY idx_campaign_template (event_id, template_id)

KEY idx_campaign_schedule (campaign_status, scheduled_at)

KEY idx_campaign_event_type (event_id, campaign_type)

KEY idx_campaign_deleted (event_id, deleted_at)

Relationships / foreign keys

event_id -> events

(event_id,template_id)->templates composite

Implementation notes

Campaign definition freezes once recipient resolution begins.

PDM-015 - eventflow_messages

One logical communication to one recipient with rendered historical snapshot.

Column	MySQL Type	Null	Default	Purpose
message_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
campaign_id	BIGINT UNSIGNED	YES	NULL	Parent Campaign
invitation_id	BIGINT UNSIGNED	YES	NULL	Invitation recipient context
attendee_id	BIGINT UNSIGNED	YES	NULL	Attendee recipient context
channel	VARCHAR(32)	NO	-	Delivery channel
recipient_name	VARCHAR(190)	YES	NULL	Historical recipient name
recipient_address	VARCHAR(500)	NO	-	Historical destination
recipient_address_normalized	VARCHAR(190)	YES	NULL	Canonical destination
subject	VARCHAR(500)	YES	NULL	Rendered email subject
rendered_content	LONGTEXT	NO	-	Exact rendered content
plain_text_content	LONGTEXT	YES	NULL	Plain-text snapshot
delivery_status	VARCHAR(32)	NO	'draft'	Message delivery lifecycle
idempotency_key	CHAR(64)	NO	-	Logical-send idempotency key
provider	VARCHAR(64)	YES	NULL	Selected provider adapter
provider_message_id	VARCHAR(190)	YES	NULL	Primary provider correlation
attempt_count	SMALLINT UNSIGNED	NO	0	Derived attempt count
queued_at	DATETIME	YES	NULL	UTC queue time
processing_at	DATETIME	YES	NULL	UTC worker start
provider_accepted_at	DATETIME	YES	NULL	UTC provider acceptance
delivered_at	DATETIME	YES	NULL	UTC delivery confirmation
failed_at	DATETIME	YES	NULL	UTC final failure
bounced_at	DATETIME	YES	NULL	UTC bounce
cancelled_at	DATETIME	YES	NULL	UTC cancellation
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC update time

Indexes and uniqueness

PRIMARY KEY (message_id)

UNIQUE KEY uq_message_idempotency (idempotency_key)

UNIQUE KEY uq_message_event_id (event_id, message_id)

KEY idx_message_event_status (event_id, delivery_status)

KEY idx_message_campaign (event_id, campaign_id, delivery_status)

KEY idx_message_invitation (event_id, invitation_id)

KEY idx_message_attendee (event_id, attendee_id)

KEY idx_message_recipient (event_id, recipient_address_normalized)

KEY idx_message_provider_id (provider, provider_message_id)

KEY idx_message_queue (delivery_status, queued_at)

Relationships / foreign keys

event_id -> events

(event_id,campaign_id)->campaigns

(event_id,invitation_id)->invitations

(event_id,attendee_id)->attendees

Implementation notes

Recipient and rendered content become immutable once processing begins.

PDM-016 - eventflow_message_delivery_attempts

Append-oriented provider transmission attempts for a Message.

Column	MySQL Type	Null	Default	Purpose
delivery_attempt_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
message_id	BIGINT UNSIGNED	NO	-	Parent Message
attempt_number	SMALLINT UNSIGNED	NO	-	Sequential attempt number
provider	VARCHAR(64)	NO	-	Provider adapter
attempt_status	VARCHAR(32)	NO	'started'	started/provider_accepted/ failed/timed_out/cancelled
provider_message_id	VARCHAR(190)	YES	NULL	Provider message reference
provider_request_id	VARCHAR(190)	YES	NULL	Provider request correlation
response_code	VARCHAR(64)	YES	NULL	HTTP/provider response code
error_code	VARCHAR(190)	YES	NULL	Sanitized error code
error_message	TEXT	YES	NULL	Sanitized error summary
attempted_at	DATETIME	NO	-	UTC attempt start
provider_accepted_at	DATETIME	YES	NULL	UTC provider acceptance
completed_at	DATETIME	YES	NULL	UTC completion
created_at	DATETIME	NO	-	UTC persistence time

Indexes and uniqueness

PRIMARY KEY (delivery_attempt_id)

UNIQUE KEY uq_delivery_attempt_event_id (event_id, delivery_attempt_id)

UNIQUE KEY uq_delivery_attempt_number (message_id, attempt_number)

KEY idx_delivery_attempt_message (event_id, message_id, attempted_at)

KEY idx_delivery_attempt_event_status (event_id, attempt_status)

KEY idx_delivery_attempt_provider_id (provider, provider_message_id)

KEY idx_delivery_attempt_request_id (provider, provider_request_id)

KEY idx_delivery_attempt_event_time (event_id, attempted_at)

Relationships / foreign keys

event_id -> events

(event_id,message_id)->messages composite

PDM-017 / 017A - eventflow_provider_events

Asynchronous provider-originated communication events with mandatory EventFlow deduplication.

Column	MySQL Type	Null	Default	Purpose
provider_event_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
message_id	BIGINT UNSIGNED	NO	-	Related Message
delivery_attempt_id	BIGINT UNSIGNED	YES	NULL	Related attempt when identifiable
provider	VARCHAR(64)	NO	-	Provider adapter
provider_event_key	VARCHAR(190)	YES	NULL	Provider-native event ID when available
event_dedupe_key	CHAR(64)	NO	-	Mandatory EventFlow SHA-256 dedupe key
provider_event_type	VARCHAR(100)	NO	-	Provider-specific event type
normalized_event_type	VARCHAR(64)	NO	-	EventFlow normalized event
provider_status	VARCHAR(100)	YES	NULL	Provider status detail
reason_code	VARCHAR(190)	YES	NULL	Structured reason code
reason_message	TEXT	YES	NULL	Sanitized reason
provider_occurred_at	DATETIME	YES	NULL	Provider occurrence time
received_at	DATETIME	NO	-	UTC EventFlow receipt time
processed_at	DATETIME	YES	NULL	UTC processing completion
processing_status	VARCHAR(32)	NO	'pending'	pending/processing/processed/failed/ignored
created_at	DATETIME	NO	-	UTC persistence time

Indexes and uniqueness

PRIMARY KEY (provider_event_id)

UNIQUE KEY uq_provider_event_dedupe (provider, event_dedupe_key)

KEY idx_provider_event_key (provider, provider_event_key)

KEY idx_provider_event_message (event_id, message_id, provider_occurred_at)

KEY idx_provider_event_attempt (event_id, delivery_attempt_id)

KEY idx_provider_event_event_type (event_id, normalized_event_type)

KEY idx_provider_event_processing (processing_status, received_at)

KEY idx_provider_event_received (event_id, received_at)

Relationships / foreign keys

event_id -> events

(event_id,message_id)->messages

(event_id,delivery_attempt_id)->delivery_attempts

PDM-018 - eventflow_audit_logs

Append-only business and security history for material EventFlow operations.

Column	MySQL Type	Null	Default	Purpose
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audit_log_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	YES	NULL	Event scope where applicable
actor_type	VARCHAR(32)	NO	-	user/guest/system/background_job/webhook/migration
actor_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
actor_reference	VARCHAR(190)	YES	NULL	Non-user actor reference
action_type	VARCHAR(100)	NO	-	Business action
entity_type	VARCHAR(64)	NO	-	Affected entity type
entity_id	BIGINT UNSIGNED	YES	NULL	Polymorphic entity ID
operation_id	CHAR(36)	YES	NULL	Business operation correlation
correlation_id	VARCHAR(100)	YES	NULL	Technical request/job correlation
change_summary	VARCHAR(500)	YES	NULL	Human-readable summary
before_data	JSON	YES	NULL	Relevant pre-change fields only
after_data	JSON	YES	NULL	Relevant post-change fields only
source_type	VARCHAR(32)	NO	'application'	admin_ui/guest_portal/rest_api/background_job/webhook/migration/system
ip_address	VARCHAR(45)	YES	NULL	Optional security context
user_agent	VARCHAR(500)	YES	NULL	Optional client context
reason	VARCHAR(500)	YES	NULL	Business reason
occurred_at	DATETIME	NO	-	UTC business occurrence
created_at	DATETIME	NO	-	UTC persistence time

Indexes and uniqueness

PRIMARY KEY (audit_log_id)

KEY idx_audit_event_time (event_id, occurred_at)

KEY idx_audit_entity (event_id, entity_type, entity_id, occurred_at)

KEY idx_audit_actor (actor_type, actor_user_id, occurred_at)

KEY idx_audit_action (event_id, action_type, occurred_at)

KEY idx_audit_operation (operation_id)

KEY idx_audit_correlation (correlation_id)

KEY idx_audit_source (source_type, occurred_at)

Relationships / foreign keys

event_id -> events RESTRICT when non-null

Implementation notes

No updated_at or deleted_at; secrets/raw credentials must never be recorded.

PDM-019 - eventflow_event_venue_snapshots

Event-specific historical venue truth captured when location is operationally locked.

Column	MySQL Type	Null	Default	Purpose
venue_snapshot_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Event
venue_id	BIGINT UNSIGNED	YES	NULL	Source reusable Venue

venue_name	VARCHAR(190)	NO	-	Historical venue name
address_line_1	VARCHAR(190)	YES	NULL	Historical address
address_line_2	VARCHAR(190)	YES	NULL	Historical address
city	VARCHAR(120)	YES	NULL	Historical city
region	VARCHAR(120)	YES	NULL	Historical region
postal_code	VARCHAR(32)	YES	NULL	Historical postal code
country_code	CHAR(2)	YES	NULL	Historical country code
latitude	DECIMAL(10,7)	YES	NULL	Historical latitude
longitude	DECIMAL(10,7)	YES	NULL	Historical longitude
phone	VARCHAR(40)	YES	NULL	Historical venue phone
email	VARCHAR(190)	YES	NULL	Historical venue email
website_url	VARCHAR(500)	YES	NULL	Historical venue website
snapshot_reason	VARCHAR(64)	NO	-	event_activated/ event_published/ manual_lock/migration/ legacy_import
snapshot_at	DATETIME	NO	-	UTC snapshot time
created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
created_at	DATETIME	NO	-	UTC persistence time

Indexes and uniqueness

PRIMARY KEY (venue_snapshot_id)

UNIQUE KEY uq_venue_snapshot_event (event_id)

KEY idx_venue_snapshot_venue (venue_id)

KEY idx_venue_snapshot_location (country_code, region, city)

KEY idx_venue_snapshot_time (snapshot_at)

Relationships / foreign keys

event_id -> events RESTRICT

venue_id -> venues RESTRICT

PDM-020 - eventflow_schema_migrations

Authoritative history for schema and controlled data migrations.

Column	MySQL Type	Null	Default	Purpose
migration_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
migration_key	VARCHAR(100)	NO	-	Stable unique migration identifier
migration_version	VARCHAR(32)	NO	-	Application/schema milestone
migration_type	VARCHAR(32)	NO	'schema'	schema/data/mixed
migration_status	VARCHAR(32)	NO	'pending'	pending/running/ completed/failed/ rolled_back/ partially_completed
checksum	CHAR(64)	NO	-	SHA-256 migration checksum
description	VARCHAR(500)	NO	-	Human-readable purpose
started_at	DATETIME	YES	NULL	UTC start
completed_at	DATETIME	YES	NULL	UTC completion
failed_at	DATETIME	YES	NULL	UTC failure
duration_ms	BIGINT UNSIGNED	YES	NULL	Duration milliseconds
executed_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress operator
execution_source	VARCHAR(32)	NO	'system'	plugin_upgrade/cli/ admin_ui/deployment/ system

from_schema_version	VARCHAR(32)	YES	NULL	Previous schema version
to_schema_version	VARCHAR(32)	YES	NULL	Resulting schema version
records_examined	BIGINT UNSIGNED	YES	NULL	Optional reconciliation count
records_changed	BIGINT UNSIGNED	YES	NULL	Optional reconciliation count
records_failed	BIGINT UNSIGNED	YES	NULL	Optional reconciliation count
validation_status	VARCHAR(32)	YES	NULL	pending/passed/failed/warning
rollback_available	TINYINT(1)	NO	0	Automated rollback available
rollback_reference	VARCHAR(190)	YES	NULL	Recovery plan reference
error_code	VARCHAR(190)	YES	NULL	Sanitized failure code
error_message	TEXT	YES	NULL	Sanitized failure summary
created_at	DATETIME	NO	-	UTC record creation

Indexes and uniqueness

PRIMARY KEY (migration_id)

UNIQUE KEY uq_schema_migration_key (migration_key)

KEY idx_migration_status (migration_status, started_at)

KEY idx_migration_version (migration_version)

KEY idx_migration_schema_version (to_schema_version)

KEY idx_migration_type (migration_type, migration_status)

KEY idx_migration_validation (validation_status, completed_at)

Implementation notes

Platform-scoped; completed migration records are immutable history.

PDM-021A - eventflow_import_jobs

Parent record for one staged CSV/XLSX import operation.

Column	MySQL Type	Null	Default	Purpose
import_job_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Owning Event
import_type	VARCHAR(64)	NO	'invitations'	Import domain
import_status	VARCHAR(32)	NO	'uploaded'	Import lifecycle
source_filename	VARCHAR(255)	NO	-	Original file name
source_file_hash	CHAR(64)	YES	NULL	SHA-256 source file hash
source_media_id	BIGINT UNSIGNED	YES	NULL	Optional WordPress media reference
mapping_definition	JSON	YES	NULL	Source-to-domain column mapping
options_definition	JSON	YES	NULL	Import options
total_rows	INT UNSIGNED	NO	0	Parsed count
valid_rows	INT UNSIGNED	NO	0	Valid count
warning_rows	INT UNSIGNED	NO	0	Warning count
invalid_rows	INT UNSIGNED	NO	0	Invalid count
applied_rows	INT UNSIGNED	NO	0	Applied count
skipped_rows	INT UNSIGNED	NO	0	Skipped count
failed_rows	INT UNSIGNED	NO	0	Failed count
uploaded_at	DATETIME	NO	-	UTC upload time
validated_at	DATETIME	YES	NULL	UTC validation completion
applied_at	DATETIME	YES	NULL	UTC apply completion
completed_at	DATETIME	YES	NULL	UTC overall completion

created_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress user reference
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC update time

Indexes and uniqueness

PRIMARY KEY (import_job_id)

UNIQUE KEY uq_import_job_event_id (event_id, import_job_id)

KEY idx_import_job_event (event_id, created_at)

KEY idx_import_job_status (event_id, import_status)

KEY idx_import_job_file_hash (event_id, source_file_hash)

Relationships / foreign keys

event_id -> events RESTRICT

PDM-021B - eventflow_import_rows

Staged, normalized and validated source rows before production apply.

Column	MySQL Type	Null	Default	Purpose
import_row_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
import_job_id	BIGINT UNSIGNED	NO	-	Parent Import Job
event_id	BIGINT UNSIGNED	NO	-	Owning Event
source_row_number	INT UNSIGNED	NO	-	Source spreadsheet row number
raw_data	JSON	NO	-	Original parsed row
normalized_data	JSON	YES	NULL	Canonical EventFlow values
row_status	VARCHAR(32)	NO	'pending'	pending/valid/warning/invalid/ready/applied/skipped/failed
validation_errors	JSON	YES	NULL	Blocking validation errors
validation_warnings	JSON	YES	NULL	Non-blocking warnings
duplicate_match_type	VARCHAR(32)	YES	NULL	exact/probable/possible/none
matched_invitation_id	BIGINT UNSIGNED	YES	NULL	Existing Invitation candidate
apply_action	VARCHAR(32)	YES	NULL	create/update/skip/reject
applied_invitation_id	BIGINT UNSIGNED	YES	NULL	Resulting Invitation
applied_at	DATETIME	YES	NULL	UTC apply time
created_at	DATETIME	NO	-	UTC staging creation
updated_at	DATETIME	NO	-	UTC update time

Indexes and uniqueness

PRIMARY KEY (import_row_id)

UNIQUE KEY uq_import_source_row (event_id, import_job_id, source_row_number)

KEY idx_import_row_status (event_id, import_job_id, row_status)

KEY idx_import_row_event (event_id)

KEY idx_import_row_match (event_id, matched_invitation_id)

KEY idx_import_row_applied (event_id, applied_invitation_id)

Relationships / foreign keys

event_id -> events

(event_id,import_job_id)->import_jobs

(event_id,matched_invitation_id)->invitations

(event_id,applied_invitation_id)->invitations

Implementation notes

Detailed staging rows are subject to controlled retention after reconciliation.

PDM-022 - eventflow_event_memberships

Event-scoped authorization between WordPress users and Events.

Column	MySQL Type	Null	Default	Purpose
event_membership_id	BIGINT UNSIGNED	NO	AUTO_INCREMENT	Internal primary key
event_id	BIGINT UNSIGNED	NO	-	Event
user_id	BIGINT UNSIGNED	NO	-	WordPress user ID
event_role	VARCHAR(32)	NO	-	owner/organizer/ coordinator/reception/ reporting
membership_status	VARCHAR(32)	NO	'active'	invited/active/suspended/ revoked/expired
is_primary_owner	TINYINT(1)	NO	0	Primary Event owner flag
granted_by_user_id	BIGINT UNSIGNED	YES	NULL	WordPress granting user
granted_at	DATETIME	NO	-	UTC access grant
revoked_at	DATETIME	YES	NULL	UTC revocation
expires_at	DATETIME	YES	NULL	Optional temporary-access expiry
created_at	DATETIME	NO	-	UTC creation time
updated_at	DATETIME	NO	-	UTC update time

Indexes and uniqueness

PRIMARY KEY (event_membership_id)

UNIQUE KEY uq_event_membership_user (event_id, user_id)

KEY idx_membership_user_status (user_id, membership_status)

KEY idx_membership_event_role (event_id, event_role, membership_status)

KEY idx_membership_event_owner (event_id, is_primary_owner, membership_status)

KEY idx_membership_expiry (membership_status, expires_at)

Relationships / foreign keys

event_id -> events RESTRICT

Implementation notes

WordPress authenticates users; EventFlow membership provides ordinary Event-level authorization.

6. Cross-Entity Business Invariants

Every operational record belongs to exactly one Event, directly or through an unambiguous parent. High-traffic records carry event_id directly.

Active Attendee count must not exceed Invitation capacity; an Invitation has at most one active primary Attendee.

One active seating assignment per Attendee and one active Attendee per Seat are enforced transactionally by SeatingService. Manual placements override routine automatic reseating.

Check-ins are append-only actions. Corrections create explicit reversal records.

Published communication template versions are immutable; campaign definitions freeze when recipient resolution begins; Message recipient/rendered payload freezes when processing begins.

Provider webhook processing is idempotent using provider + event_dedupe_key.

WordPress user/media references are application-enforced rather than hard foreign keys to WordPress core tables.

7. Import and Migration Rules

Imports use staged import_jobs and import_rows persistence. Parsing, normalization, validation, duplicate detection and organizer review occur before production apply.

Schema/data evolution uses schema_migrations with stable migration keys, SHA-256 checksums, validation outcomes, reconciliation metrics and rollback/recovery references.

8. Database Platform and Portability

EventFlow v1.x requires MySQL 8.0+ or MariaDB 10.11+, InnoDB, utf8mb4, transactions and database-enforced foreign keys.

Logical JSON fields must remain portable across supported MySQL/MariaDB engines and must not depend on MySQL-specific binary JSON semantics/operators/indexes without a future ADR.

Installation and upgrades perform environment readiness checks before applying migrations.

9. Baseline and Change Control

The authoritative draft DDL is sql/eventflow-schema-baseline-v0.9.sql. The {prefix} placeholder is replaced with the active WordPress database prefix at migration runtime.

Acceptance of EF-DOC-005 v1.0 will freeze the Sprint 3 database design for implementation. Subsequent material schema changes require traceability to requirements/ADRs/design and controlled migrations.